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Mineral Wealth of Cuddapah Basin and its Use for Sustainable Development-An Overview

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Abstract

India is endowed with a rich variety of mineral resources due to varied geologic and tectonic milieu. To ensure sustainable development of mineral industry, in place of ill oriented approaches, it is essential to have a properly estimated resource and reserve statistics. This can be achieved if a pragmatic exploration strategy is adapted. One of the important segments known for mineral wealth is Cuddapah basin. Cuddapah basin, endowed with rich mineral wealth, is one of the important and fairly well studied geological units in peninsular India. Cuddapah basin has been well known for a variety of mineral resources, such as barytes, asbestos, copper, iron, fullerene, lead, diamonds, uranium, gold, silver, limestone, clay, quartz, talc-steatite, granites besides Cuddapah slabs. Recently it is also established that the basin`s hydrocarbon potential is good. Effective exploration strategy coupled with environmentally viable mineral specific mining and product development strategies need to be developed and used to have organized mineral industries, in and around Cuddapah basin, many parts of which are underdeveloped economically. In this write up specific details pertaining to source of a particular mineral from available details, implemented exploration strategy, statistics of overall resource and reserve details are listed. Organized strategy for safe mining practices and steps/ procedures to be adapted in establishing mineral industries, for sustainable development of the region are also detailed.

Key words: *Cuddapah basin, mineral resources, mining, sustainable development*

Introduction

Mining is a major economic activity in India (UNEP1997). The Ministry of Mines (MOM), Government of India is responsible for the entire minerals and mining sector in the country that includes legislation, administration, policy formulation etc. in respect of all mines and minerals other than coal and lignite, natural gas and petroleum, but including offshore minerals. In India, the minerals are classified as minor minerals and major minerals. All the mineral legislations in the country conform to the provisions of the MMDR Act, 1957. MOM through its attached office, Geological Survey of India (GSI) facilitates exploration, geological mapping and mineral resource assessment in the country. Indian Bureau of Mines (IBM), a subordinate office of the MOM is mainly responsible for regulation of mining in the country. Government of India has recently enacted amendments to Mines and Minerals (Development & Regulation) Act, 1957 (MMDR) and notified rules that would help in overcoming many challenges associated with the minerals and mining sector such as low level mineral exploration and exploitation, low technology deployment, fragmented and small concession areas etc. India is poised to witness great leaps of growth in minerals and mining sector (Ministry of Mines, 2015). Most of the exploration activities in the country are of conventional type with restricted input from geology, geochemistry, geophysics and remote sensing. The finds so far are located near the surface (mostly up to a vertical depth of 100 to 200 m). Therefore, with fast depletion of easily accessible and shallow or near surface ore bodies and decline in the rate of locating new mineral deposits within shallow depths, the challenge lies in identifying new area for locating near surface deposits and “deep seated” and “concealed/hidden” ore bodies through modern and sophisticated exploration methods/techniques on the basis of conceptual studies (Ministry of Mines, 2015).

The Middle- Upper Proterozoic Cuddapah basin, has been well known for a variety of mineral resources, such as baryte, asbestos, uranium, diamonds, gold, silver, copper and lead, besides limestone, steatite-talc, quartz and Cuddapah slabs. It is also recently established through focused integrated geological, geochemical and geophysical investigations that the basin`s hydrocarbon potential is good.

The Cuddapah is an epicratonic Proterozoic basin situated in the state of Andhra Pradesh, South India. It is 440 km long and has a maximum width of 200 km in the middle. The basin covers an area of 44,500 km2 with the aggregate stratigraphic thickness of 12,000 m (Kale, 1991). The sediment thickness increases from west to east signifying the deepening of basin towards east and sediments are dominantly composed of argillaceous and arenaceous sequences with subordinate calcareous successions. Lithostratigraphically, the basin is divided into lower Cuddapah Supergroup and upper Kurnool Group. The Cuddapah Supergroup is overlain by the Kurnool group with an unconformity (Ramam and Murty, 1997).The Cuddapah Supergroup of middle Proterozoic (1600–1300 Ma) age is divided into Papagni (2100 m), Chitravathi (6000 m) and Nallamalai (3500 m) groups, composed dominantly of argillaceous and arenaceous sediments. Each of the three subgroups starts with quartzite and ends with a shale unit representing cyclic repetition of quartzite and shale sequence, reflective of transgression and regression in an episodically sinking basin. Contemporaneous igneous activities are manifested as sills, flows and other intrusives along the western periphery of the basin and the eastern Nallamalai group (Nagaraja Rao et al., 1987). The Cuddapah Basin has progressively evolved through the formation of a series of sub-basins, viz., the Papagni and Chitravathi Groups in the western sub-basin, Nallamalai sub-basin in the eastern sub-basin, Srisailam Quartzite in the Srisailam sub-basin and the Kurnool and Palnad Groups in the Kurnool/Palnad sub basins. The Basin hosts a wide spectrum of rich mineral resources.

It is clearly pointed out by Kaila (1979), Reddy et al (2004) and Reddy et al (2010) through crustal velocity structure along Kavali-Udipi Deep seismic sounding profile that entire basin from east to west could possibly has significant structural features that reflect uniqueness of the basin, from natural resources point of view. The regionally extending prominent thrust noticed at the eastern flank of the basin that has a double Mohorovicic discounuity (Moho) acts as a feeder for mineral rich high density deeper formations coming up to near surface depths. The known mineral zones immediately east of this mega thrust at Vinjamoor, Nellore district and the clear structural changes from Badvel to regions beyond Mydukuru (within the basin), followed by mega up doming of high density formations at the western flank of the basin at Duttaluru, apart from in between prominent structures associated with Kurnool group of rocks suggest that there is every possibility every segment of this basin has rich mineral wealth. Till now its mineral wealth has not been exploited in a judicious way due to ill organised mining practices at random locations.

Suggested strategy for organized development of mineral industries in Cuddapah basin

In view of paucity of organized exploration and exploitation strategy, especially during the last 3 to 4 decades, the excellent works carried out by number of pioneers starting from King remained as unutilized treatises. Our conjectured opinion that all parts of the basin could possibly contain rich quantities of minerals can be substantiated or negated only through detailed exploration of entire basin. While planning such a large quantum of work a group of experts (from different branches of earth science, belonging to GSI, NGRI, AMD, NMDC, and universities) can make use of available geological, geophysical, geochemical and geochronological results/ models as base and suggest viable exploration studies covering strategic locales of the basin, to extend potential zones laterally and vertically, for organized exploitation.

As of now individual mineral resource and reserve statistics are worked out using available scanty exploration and exploitation in puts. From a practical point of view, such a procedure is unsatisfactory .It needs to be kept in mind that a resource is a concentration of a naturally occurring material (solid, liquid, or gas) in or on the crust of the earth in a form that economical extraction is presently feasible. A reserve is that portion of the total resource, which is identified and from which usable materials can be economically and legally extracted at the time of evaluation (US Geological Survey, 1975). Since resources include both identified reserves and other materials that either are identified but not presently economically or legally available or are undiscovered, the statistical details are subjective in general as many parts of a potentially mineral rich region have not yet been scanned thoroughly to have specific statistics. Keeping this in view we have structured present manuscript with available inputs. We do, however, know the statistical details need to be reevaluated after completion of close grid integrated high resolution exploration exercises.

Mineral wise details

We do hope our present initiative acts as a base to build effective strategies. This write up is structured to bring out a broad over view of presently available information , information pertaining to gaps in our understanding of the rich heritage of the basin , followed by an outline of suggested strategy to bridge these gaps. Details pertaining to type of occurrence, statistics of reserves, demand and present adopted policy to meet the demand, mining options, probable approaches for specific mineral based industry development are given below, while restricting the length of presentation to meet IJEE norm. Once the suggested strategy is approved for adaptation quality control mechanism, mid-course correction approaches could be developed with inputs from different sources. Mineral specific details are given below of majority of minerals. They clearly show that Cuddapah basin has rich mineral wealth and these minerals can be mined and used to develop in-situ mineral industries, following various established norms.

Asbestos

Asbestos belongs to a group of six naturally occurring fibrous silicate minerals. The physical properties, besides fibrous character, such as, fineness, flexibility, tensile strength & length of fibres, infusibility, low heat conductivity and high resistance to electricity & sound as also to corrosion by acids, make asbestos commercially important. Chrysotile asbestos is the most important commercial variety of asbestos for spinning into yarn and woven into fabrics based on properties like fibrous nature, spin ability, incombustibility, resistance to heat, acids and alkalies, filtration property, significant strength and property to absorb cements. Commercially, chrysotile asbestos is far superior in physical properties and hence more valuable than amphibole asbestos.

Chrysotile asbestos occurs in the zones of serpentinization between dolomite sills and magnesium rich limestone invariably at the upper contact, of Vempalli Formation of Papaghni Group of Cuddapah Super group along 15 km long belt from Brahmanapalli–Lopatnutala of Pulivendula Mandal, Kadapa district. It occurs as a few millimeters to about 10 cm long fibers from near surface to a depth of 200 m (Deb 1980).It is estimated that reserve of 2.5 lakh tonnes are available in this patch of Cuddapah basin (Deb et al., 1991; www. nsdl.niscair.res.in).

It can be extracted through underground mining, as it is developed along the inclines that are sub-parallel to the bedding planes. The other occurrence of asbestos is at the upper contact of dolerite sill in Vempalli dolomite near Rajupalem (Sunitha V. et al., 2014). As per the UNFC (United Nations International Framework) system, the total resources of asbestos in the country as on 1.4.2010 are placed at 22.17 million tones. Of these, 2.5 million tonnes are reserves and 19.6 million tonnes are remaining resources, including those available from Cuddapah basin (Indian Minerals Year Book: 2013: Asbestos). Since the demand is considerable, as of now India's asbestos requirement is mainly met through imports from Russia, Kazakhstan, Brazil and China (Indian minerals: Year Book: 2013: Asbestos). As such it is essential to revitalize the Asbestos mining and Asbestos Industry, by introducing pollution free mining and product development practices to arrest probable health related problems.

Barytes

Barytes or barites is the moderately soft crystalline mineral form of barium sulphate. Approximately, 80% barytes produced worldwide is used for oil and gas drilling as a weighting agent (drilling mud) because of its unique physical and chemical properties like heaviness, high specific gravity and magnetic neutrality. This principal mineral of barium is also used as a feedstock for producing various barium compounds. It is also utilised as filler, extender and aggregate (Indian Minerals Year Book: 2013: Barytes). Barytes after converting into barium carbonate is used in the manufacture of ceramics and glass. Andhra Pradesh is the leading producer of barytes and contributes about 94% of the total country's production. The Mangampet deposit in Kodur Mandal of Kadapa district of Andhra Pradesh, a prominent bedded deposit, is the single largest barytes deposit in the world, with a resource of 37 Mt and an estimated reserve of 70 million tonnes (Neelakantam, 1987).Andhra Pradesh Mineral Development Corporation (APMDC) exploiting the Mangampeta deposit under public sector is producing about 5 to 6 lakh tonnes per annum. It intends to increase the production. Huge reserves of barytes are found in Kadapa, Prakasam and Nellore districts. Mangampeta barytes with grade 92% $BaSO_4$and an average specific gravity of 4.21 is used in oil & gas well drilling. The other grades are catering the needs of Chemical and Paint industries. Barytes also occurs in Vempalli dolomite and associated basic igneous rocks as veins in Pulivendula, Kamalapuram and Kadapa taluks. It is estimated that, apart from Mangampeta, there are reserves of the order of about 7 lakh tonnes of barytes of all grades within depth of 30m from the surface in the mineralized belt extending for about 100km form Velidandla in the west to Mittamindapalle in the east of Kadapa district. Apart from Kadapa district important occurrences of Barytes are also found near Hussainapuram, Valasala, Chinnamalkapuram, Blapalapalli, Ahobilam and Sirvel villages of Kurnool district. Vein barites at the contact of basic skills with Tadpatri Shale are reported from Peddapaya and Boyanapalle of Kurnool district. Near Chinna Ahobilam of Kurnool district barytes veins are located along fractures/shears within Bairekonda Quartzite (www.kurnool.ap.gov.in). The total resources of barytes in India as on 1.4.2010 as per UNFC (United Nations International Framework)system are placed at 73 million tonnes constituting 43% reserves and 57% remaining resources. By grades, 40% resources are of oil-well drilling grade followed by 6% of chemical grade, 1% of paint grade and 33% of low grade. About 20% resources are of other, unclassified and not-known categories. Barytes can be extracted using open cast mining practices. Since barytes can be used in preparing barium chemicals and also as basic filler in paints industry an organized industrial development can be achieved, by properly enhancing quality norms, including introduction of a sequence of automatic processing techniques.

Uranium

Uranium exploration in the Cuddapah basin was initiated in the late 1950's to search the quartz-pebble-conglomerate type uranium mineralization, which had dominated the world uranium supply at that time. However, the basal Gulcheru conglomerates at the base of Cuddapah basin were found to be thoriferous. Subsequent exploration in the late 1980's brought out significant uranium mineralization in dolostone. A significant breakthrough was achieved in early 1990's when uranium mineralization was located along the unconformity between Srisailam Formation of Cuddapah Super Group and the basement granites, thereby establishing in India, for the first time, the presence of unconformity related uranium mineralization - a category considered most potential world over (Sinha et al., 1995). The systematic and intensive exploration programme conducted by the Atomic Minerals Directorate for Exploration and Research (AMD) within the Cuddapah basin led to the recognition of three distinct types of uranium mineralization, viz., 1. Strata bound, 2. Fracture controlled (both basement granite and sediment hosted) and 3. Unconformity -related. Out of this unconformity-related uranium occurrences are in close proximity with the unconformity between the Srisailam quartzite/ Banganapalle quartzite and the basement granite.

Strata bound Uranium mineralisation

The strata bound uranium-mineralisation in southwestern part of the Cuddapah basin is unique in the sense that no such strata bound uranium deposit hosted by carbonate rocks is reported in the world. Uranium mineralisation is hosted by impure phosphatic dolostone of the Vempalle Formation of the Papagni Group. It extends from Chelumpalli in the northwest to Maddimadugu in the southeast over a belt of 160 km with promising mineralisation at Tummalapalle, Racha kuntapalli and Gadankipalli in the central part of the basin. Mineralisation occurs along the bedding plane, mainly in the form of ultra-fine vanadium-bearing pitchblende and U-Si-Ti complex. An essentially syngenetic origin is contemplated for this dolostone hosted mineralisation. A deposit containing over 15,000 tonnes of U3O8 has been estimated for the explored blocks at Tummalapalle, Rachakuntapalle and Gadankipalle and further potential exists in this setting. Since these deposits are of low grade (0.042% U3O8) and associated with dolostone, their exploitation depends on the development of economically viable carbonate leaching technology.

Fracture/Shear controlled uranium mineralization

Fracture controlled uranium mineralisation is both basement granite hosted as well as sediment hosted (Gulcheru Formation, the oldest member of the Cuddapah Super Group) and occurs along the southern margin of the Cuddapah basin.

(a) Basement granite-hosted fracture controlled:

Rayachoti – LakkireddiPalli of Kadapa district has over 50 uranium-mineralized fracture zones in desilicified and deformed granitoid. The exploration of these zones has led to identification of fracture controlled, lensoid hydrothermal uranium mineralisation (Sinha, 1995). The intensely fractured area between Mulapalli and Konampeta of Kadapa district holds promise for locating deposits of this type.

(b) Sediment (Gulcheru Formation)-hosted fracture-controlled:

Fractured Gulcheru quartzites have indicated high uranium contents (up to 1.00% U3O8) at several localities spread over 40 km stretch between Madyala Bodu to west of Racha-kuntapalle (Jayagopal et al., 1996).

c) Unconformity-related Uranium mineralisation in Cuddapah basin:

Workable deposits of unconformity-related type have been established at Lambapur-Peddagattu, Nalgonda district (Telangana state) and Koppunuru, Guntur district in Andhra Pradesh, along NW margin of the Cuddapah basin. Recent investigation in the northern part of Palnad sub-basin has resulted in locating uranium mineralisation in Rallavagu Tanda. In this area, mineralisation occurs along the unconformity between the basement granite and the Banganapalle quartzite. Pitchblende and coffinite are the main uranium minerals and associated sulphide minerals include pyrite, chalcopyrite and galena (Sinha et al., 1994).

The large limestone hosted uranium mineralisation under development at Tummalapalle in south -eastern part of Cuddapah basin in Kadapa district holds promise for multifold increase in uranium production in future (Gupta et al., 2010). At Tummalapalle, the U-mine is the most modern underground mine and the processing plant is fully mechanized. This project would treat 3,000 tonnes of U-ore per day. Huge deposits of natural uranium, which promise to be one of the top 20 of the world's reserves, have been found in the Tummalapalle belt in the southern part of the Cuddapah basin. AMD has so far discovered 44,000 tonnes of natural uranium (U_3O_8) in just 15 line km of the 160-km long belt. Exploration strategy for unconformity related deposits in Cuddapah basin and exploration strategy for Uranium in SW part of the basin have been detailed by Anjan Chaki (2009).The strata-bound uranium mineralisation hosted by Vempalle dolostone accounts 23% of the Indian resources. Though the dolostone hosted Tummalapalle uranium deposit was established in the early nineties, because of techno -economic constraints, the deposit remained dormant. As a consequence of the development of an innovative pressure alkali leaching process, the deposit became economically viable and a mine and mill are being constructed there. Recent exploration inputs are leading more such strata-bound low grade uranium deposits in the extension areas of Tummalapalle (Anjan Chaki, 2010).

Diamonds

Diamond occurs in two types of deposits primarily in igneous rocks of basic or ultrabasic composition and in alluvial deposits derived from the primary sources. Its composition is pure carbon and has cubic crystal system and common form octahedron. Primary sources of diamonds are kimberlite pipes and vents, and lamproite, or peridotite dykes. Secondary source is in conglomerate beds, alluvial gravels and sand. The kimberlites are dense, and dark coloured ultrabasic rocks, rich in magnesium, containing olivine, enstatite-bronzite, chromediopside, phlogopite, and pyrope garnet with minor amount of ilmenite and perovskite. Diamonds require very high pressures for generation and growth, which is not realized in the normal crust of the earth. It is therefore believed that diamonds originate in the upper mantle or in the root zone of thickened continental crust and are brought to the crust as inclusions in kimberlite pipes (Deb, 1980). Chelima dykes represent an important phase of igneous activity in the Cuddapah Basin. These dykes are also widely regarded as a source of nearby alluvial diamonds. Recently, questions have been raised regarding their petrological status as lamproites and on the possibility of crustal contamination influencing their geochemistry. Petrology and geochemistry are indeed consistent with their nomenclature as lamproites. Crustal contamination is shown to be minimal in influencing their geochemistry. Available geochronological data make them the oldest yet recorded lamproites (ca.1400 Ma) in the world. Chelima dykes were derived from an ancient and anomalously enriched melt source region with lower-time integrated Sm/Nd ratios than Bulk Earth and evolved in isolation from the convecting mantle. Petrogenetic modelling reveals that their source regions have been strongly depleted in the garnet stability field followed by enrichment by a metasomatic melt rich in LREE and other incompatible elements before final partial melting. Despite their geographical proximity the Chelima dykes (along with those at Zangamarajupalle) are considered to be temporally as well as genetically unrelated to the (i) other igneous activities in the Cuddapah sediments and (ii) adjoining kimberlites in the Wajrakarur area towards the western margin of the Cuddapah Basin. Chelima dykes occur in an altogether different geodynamic setting compared to the deformed and metamorphosed alkaline rocks of the Eastern Ghats Mobile Belt, which are recently opined to have been located in an ancient suture zone (Chalapathi Rao N. V 2007). A number of kimberlites and lamproites were emplaced during the Proterozoic in the Eastern Dharwar Craton and Cuddapah Basin. Known kimberlite occurrences in the Dharwar Craton are confined to an area immediately west of the Cuddapah Basin. They occur as two clusters, in the Anantapur (Andhra Pradesh) and Mahabubnagar (Telangana) districts that are separated from each other by 4150 km. In contrast, lamproites are confined to the Cuddapah Basin (Chelima and Zangamarajupalle) and its NE margin (Ramannapeta and Jaggayyapeta; Reddy et al., 2000). The kimberlites and lamproites of the Indian sub-continent offer a rare opportunity to investigate Proterozoic tectono-magmatic processes operating when southern India formed a part of the Gondwana supercontinent. Recent studies shed further insight into the genesis of, and genetic relationship between, kimberlites and lamproites (Chalapathi Rao N.V et al., 2004). The Chelima and Zangamarajupalle lamproite dykes intrude slates and phyllites of the Nallamalai Group, within the Cuddapah Basin (Bhaskar Rao, 1976; Sreeramachandra Rao, 1988). Neither dyke has yet been tested for diamond by modern exploration techniques. A more interesting occurrence is from fluvial conglomerates in the Banganapalle Quartzite. Banaganapalle (located in Cuddapah basin) employed thousands of miners - centuries ago, and virtually none of the rocks have been left in situ. This occurrence indicates a Precambrian diamond source, confirmed by 1225-1140 m.y. isotopic dates of pipes. The Ramannapeta lamproite intrudes granitoid country rocks at the NE margin of the Cuddapah Basin and lies close to the alluvial workings on the banks of the Krishna River where many famous Indian diamonds, including the Koh-I-Noor, have been recovered. The primary source of these diamonds is uncertain. Earlier workers proposed the Anantapur kimberlites and Chelima lamproite as the source rock for these diamonds. However, based on the close proximity (<10 km) of the Ramannapeta lamproite to these workings and the >200 km distance of the nearest known kimberlites (Chalapathi Rao N.V et al 2003) suggest that the Ramannapeta body may be the primary source for the diamonds of the Krishna valley.

In view of the above suggestion it is essential to carry out modern exploration techniques involving geological, geochemical and geophysical techniques. Kimberlite and lamproite formations take place in peridotite and ecologite in the mantle lithosphere associated with thick cool roots of archaen continent nuclei. The presence of the root may be detected by geochemical studies of the macrocyst mantle minerals in diamondiferous kimberlite and to a lesser extent lamproites. Since the target is very evasive and probably present below thick overburden, exploration needs to be planned using high resolution dense data coverage adapting an integrate approach. The interpretation of indicator mineral compositions is being upgraded continually by the use of modern analytical techniques and data bases with a view to diamond prospecting. The exploration results culminate with drilling to understand the 3rddimension of the mineralised zones, and data processing using 2D and 3D geological softwares to define the 3D ore body model and Resource estimation. The presence of specific indicator minerals in unconsolidated sediments provides evidence of a bedrock source in the provenance region and in some cases, the chemical composition of the minerals is associated with the ore grade of the bedrock source. As few as one sand-sized grain of a particular indicator mineral in a 10 kg sample may be significant. As per the UNFC (United Nations International Framework) system as on 1.4.2010, all India resources of diamond are placed at around 31.92 million carats. Out of these, 1.04 million carats are placed under reserves category and 30.88 million carats under remaining resources category. By grades, about 2.37% resources are of gem variety, 2.63% of industrial variety and bulk of the resources (95%) are placed under unclassified category. Cuddapah basin and other resources of Andhra Pradesh account for 5.71% of Indian diamonds.

Gold

Gold is a relatively scarce metal in India. Properties of gold, which make it useful in industry, are malleability, ductility, colour, resistance to corrosion, high electrical conductivity, luster and therapeutic effects of some of its salts. India is a minor producer of gold but has huge demand in the country mainly in jewellery and ornament sector. The domestic demand is mainly met through import of gold (Indian Minerals Year Book 2013: Gold). No metal or alloy substitute has all the properties of gold, and therefore, the emphasis is only on reduction of gold content rather than substitution. There are old workings for gold in hornblende schist and amphibolites at Jonnagiri, Pathikonda of Kurnool district. M/s Geomysore services India Pvt. Ltd. has filed mining lease application for Gold and other precious minerals over an extent of 60,000 hectares in Jonnagiri- Pathikonda tract.
(Source:http://www.kurnool.ap.gov.in/departmentView.apo?mode=getDepartment&departmentFlag=MIG&subDepartmentFlag=MIG).

Base Metals

Lead-zinc deposits are generally accompanied by silver, hosted within the lead sulfide mineral galena or within the zinc sulfide mineral sphalerite. Lead and zinc deposits are formed by discharge of deep sedimentary brine onto the sea floor (termed sedimentary exhalative or SEDEX), or by replacement of limestone, in skarn deposits, some associated with submarine volcanoes (called volcanogenic massive sulfide ore deposits or VMS) or in the aureole of sub volcanic intrusions of granite. The vast majority of SEDEX lead and zinc deposits are Proterozoic in age. Copper is found in association with many other metals and deposit styles. Commonly, copper is either formed within sedimentary rocks, or associated with igneous rocks. The world's major copper deposits are formed within the granitic porphyry copper style. Copper is enriched by processes during crystallization of the granite and forms as chalcopyrite — a sulfide mineral, which is carried up with the granite. Sometimes granites erupt to surface as volcanoes, and copper mineralisation forms during this phase when the granite and volcanic rocks cool via hydrothermal circulation. Often copper is associated with gold, lead, zinc and nickel deposits.

In Cuddapah basin several small deposits of Pb-Zn or Cu are known from the Cumbum Formation and, to lesser extent, the Vempalle Formation. Individual deposits are a couple of million tonnes or less. Grades are mostly less than 1%. Deposits are generally strata-bound, but also seem to have some structural input as well. Base metal mineral deposits are known in the Cumbum Formation, Vempalle and Tadpatri Formations spread in Agnigundala, Zangamrajupalle-Varikunta, Rayavaram-Chinavani-Palle, and Gani-Kalva and Pulivendla belts of Cuddapah basin. Also, evidences of mining activity, in recent past, for copper, lead-Zinc are seen in the vicinity of Bollaplle of Vinukonda Taluk and Karempudi of Palnad taluk of Guntur district. Important deposits occur near Bandalamottu, Nallakonda, Dhukonda areas of Ipur Mandalam of Guntur dist. Hindustan Zinc Ltd started exploiting lead deposit of Bandalamottu, Copper mineralization also occurs around Gani and Kalava in the form of disseminations within the basic intrusive and Tadpatri shales. Lead-zinc mineralisation associated with copper is found in dolomite of Cumbum Formation in Chelima area. Massive Pyrite occurs in association with metavolcanics of Jonnagiri Schist Belt (http://web-grafix.in/apmines/?page_id=889).

Hydrocarbon Potential

Cuddapah basin has all the necessary prerequisites for hydrocarbon generation and accumulation, such as significant sediment thickness, favorable structures, biological life, geothermal gradient needed for maturation. The sedimentary sequence of Cuddapah Super group is divided into lower Papagni Group (2100 m), Chitravati Group (6000 m) Nallamalai Group (3500 m) and younger Kurnool Group (520 m) comprising quartzites, limestone and shale units, separated by an unconformity. The intrusive present in the basin are due to the igneous activity that occurred simultaneously during sedimentation and is thought to be a local phenomenon. All sediments in the basin are mature and un- metamorphosed except on the eastern part due to the thrusting of Eastern Ghat Mobile Belt. The Vempalle formation of Papagni Group and Tadpatri Formation of Chitravati Group are characterized by the presence of stromatolites of columnar structure demonstrating the extent of proliferation of microbial life during Cuddapah sedimentation. In the southwest part of the Cuddapah basin (Thummalapalle Gandankipalle area) the shales contain thin stringers of brown colored organic matter and humic acid produced by decay of algal colonies.

The northwestern and southeastern regions of the basin have heat flow over 130 mWm^2 and these regions are considered to associate with highly prospective zone for the generation of hydrocarbons. The thickness of the sediments where the Nandyal shale (Upper Kurnool Group) exposed, is in the range of 6000-9000 m and these shales are considered as a future target for petroleum exploration. The Vempalle limestone/dolomite can act as good reservoir rocks with its vuggy and fracture porosity. The Papagni and Chitravati Groups are mostly undisturbed and have the prerequisites, which can be favorable for generation and accumulation of hydrocarbons. The eastern margin of the basin (Nallamalai Group) is highly disturbed with isoclinal and recumbent folds, complex faulting and also thrusting with the Eastern Ghat Mobile Belt. However, scientists have shown that these form ideal sites to look for the traps below the Nallamalai group of sediments. The structures like synclines, anticlines, fault closures etc. can play an important role in hydrocarbon generation and entrapment.. Adsorbed Soil Gas Surveys carried out by NGRI jointly with DGH have shown that the basin has hydrocarbon generation potential (Unpublished Tech. Rep. No. NGRI- 2005-Exp-499). Integrated geological, geochemical, geophysical and remote sensing studies need to be carried out to demarcate the potential hydrocarbon zones (Vishnu Vardhan, C et al., 2007).Since extraction of oil and gas from shale requires controlled fracking operations that can affect environment and groundwater resources, it is essential to carry out detailed pilot project level fracking operations in select areas to select state of the art fracking technique, to avoid setbacks to both Man and Environment. However, as our energy basket is not filled properly exploitation of thick shale beds might become essential in future.

Limestone

The term limestone is applied to any calcareous sedimentary rock consisting essentially of carbonates. The two most important constituents are calcite and dolomite. Limestone often contains magnesium carbonate, either as dolomite $CaMg(CO_3)_2$ or magnesite ($MgCO_3$) mixed with calcite. Limestones altered by dynamic or contact metamorphism become coarsely crystalline and are referred to as 'marbles' and 'crystalline limestones'. The principal use of limestone is in the Cement Industry. Other important uses are as flux in metallurgical processes; in Glass, Ceramic, Paper, Textile and Tanning Industries; for manufacture of calcium carbide, alkali and bleaching powder; for water purification and sugar refining; in fertilizer (calcium ammonium nitrate) and as soil conditioning agent in agriculture; crushed stone for ballast and filler in concrete and asphalt; as rectangular slab in lithography. As well as being a popular building stone (regular flaggy parting), the Narji Limestone is a source of both cement and blast furnace-grade lime. Flux-grade limestone is also developed secondarily from dolomite, where dolerite sills intrude Vempalle Formation of Cuddapah basin. Analysis of Aeromagnetic anomalies, from the regions of exposed basement rocks and hard rock terrains, greatly help in understanding the subsurface magnetic lithology and fracture/ fault lineament pattern. The semi-arid tracks of limestone/ shale sequence belonging to Kurnool sediments are underlain by comparatively porous quartzites of the Chitravati group of rocks of Cuddapah formation. The shallow depths obtained from aeromagnetic anomalies are confined towards the south west corner of the area and coincide with the surface exposures of the Quartzite hillocks. The WNW trending zone having a depth range of 200 mts between Kanala-Akkampalle towards west of north-south trending Kundra Vagu indicates a 200 mts thick Narji limestones overlying the Gandikota Quartzites. A gradual thickening (up to 600 mts) of the limestone/shale of Kurnools is observed in the south east corner, near and around Suddepalle. The Gandikota Quartzite horizon deepens towards south east and is overlain by the Kurnools whose thickness is around 600mts in Suddapalle area and thus coincides with the geological estimates (Butchi Babu et al., 2008).

Limestones of Kurnool district are of flagstones. Cement grade limestones are used by M/S Panyam Cements and Mineral Industries near Bugganipalli. High calcium limestones are used in industries near Dhone. Narji limstone are found in Dhone, Kurnool, Nandikotkur, Koilakunta and in several areas of the district. The Narji limestones contain both the massive and flaggy limestones. The massive deposits are used in cement manufacture while the flags are quarried as building material. Major outcrops of Narji limestones are found at Sathanikota, Bukkapuram and Bethemcherla. Koilakuntla limestones are mostly light grey calcareous flags, and massive grey limestones. These are seen at the out crop along the course of Kunderu river. They are found at Appalapuram, Varukur near Kurnool, at Nagaluti and Nandikotkur and southern parts of Sirvel. The Koilkuntla limestones are quarried extensively at places like Vyalawada and Joladursi, for use as building stones (slabs). The G.S.I. conducted detailed investigations by drilling in Narji limestones near Ankireddypalli. The investigations revealed the occurrences of both flux grade and cement grade limestones in this area. The total reserves of flux grade limestone in this area are of the order 47 million tonnes of measured category and 25 million tonnes of indicated category over an area of 4.68 sq.Km. High grade tuffaceous limestones are found at various places in Kurnool district. Calctufa is best developed in Vempalle and Narji horizons close to perennial water courses. The tuffaceous limestones are white to dirty white in colour. They are hard and porous. Large deposits of calc tufa are found near Madhavaram, Itikyala and Ramabadrapalli in Koilakuntla area, Palkur, Ramathirtham, Nandav-aram in Bugganipalli area and Bhogeswaragudi in Nandyal area. The deposits of Dhone sector are the richest in terms of quality. Buff to light coloured high grade limestones occur near Malkapuram railway station. These are hard, massive and very rich in calcium content and believed to have been formed as a result of contact effect of dolomites with dolerite sills in Vempalle formation. These limestones are being supplied to calcium carbide plant at Hagari and to other paper and chemical industries in the state and outside the state. There are 49 mining leases of which 41 are working. G.S.I. investigated these areas for similar occurrences, as this type of limestone is in great demand. They could indicate two or three probable horizons in Vempalle dolomite in Dhone sector with a probable reserve of 5 to 6 million tonnes (Source: http://web-grafix.in/apmines/?page_id=889).

The total resources of limestone of all categories and grades as per UNFC (United nations International Framework system) as on 1.4.2010 are estimated at 184, 935 million tonnes, of which 14,926 million tonnes (8%) are under reserves category and 170,009 million tonnes (92%) are under remaining resources category. Karnataka is the leading state having 28% of the total resources followed by Andhra Pradesh (20%) (Indian Minerals Year Book: 2013: Limestone).

Fullerene

Fullerene is a pure carbon molecule composed of at least 60 atoms of carbon and exhibits a Bucky ball structure. The discovery of C60, C70, the carbon cage molecules popularly known as fullerenes, named after the famous Architect, Buckminster Fullerene (Kroto et al 1985), opened new and vast vistas in understanding these curious molecules. The fullerenes, like artificial diamonds, can be synthesized from Carbon. Fullerenes, their derivatives and Carbon nano tubes have a number of interesting properties due to their unusual structures and sizes. The size of the individual fullerene molecules makes them ideal building blocks for use in designing molecular units that find application in nanotechnology. The fullerene family of carbon molecules possesses a range of unique properties. A fullerene nano-tube has tensile strength, about 20 times that of high-strength steel alloys, and a density half that of aluminum. Carbon nano-tubes demonstrate superconductive properties, for commercial applications, including computer memory, electronic wires. Five black Tuff samples from the Mangampeta area of the Cuddapah Basin, analyzed at the Stanford University (Sreedhar Murhty 2005) indicated presence of C60, C70 and C84, suggesting the presence of naturally occurring fullerenes in this part of the world. Encouraged by convincing results in Cuddapah basin and realizing the vast application potential of this very important and strategic mineral, the APMDC has come forward to support further studies in this area under a research project and entrusted the responsibility of executing the same to M/s. Geo Resources and Technologies Private Limited (GRTC). Based on the integrated geological and geophysical surveys across Mangampet Barytes mine, a model is proposed for the genesis of fullerene. The model postulates that the volcanic activity, supposed to have been responsible for the origin of barytes, played a significant role in creating suitable thermal conditions and environment conducive to transformation of the existing hydrocarbon material into higher order allotropes of carbon representing the family of fullerenes. The higher quantities of fullerene are attributed to the sampling of host rock from relatively deeper levels. Alternately the magmatic origin model envisaged for barytes may also hold good for fullerene as well, since the tuff and barytes as also the tuff and fullerene are observed to be intimately associated (http://inovativeblog.blogspot.in/2011/05/fullerenes-in-kadapa-district.html).

Talc & Steatite

Talc and Steatite are finding a great demand in the ceramic industry because they impart the required properties and the desirable qualities to the ceramic articles. Talc is a hydrous magnesium silicate. In trade parlance, talc often includes: (i) the mineral talc in the form of flakes and fibres; (ii) steatite, the massive compact cryptocrystalline variety of high-grade talc; and (iii) soapstone, the massive talcose rock containing variable talc (usually 50%), which is soft and soapy in nature. Commercial talc may contain other minerals like quartz, calcite, dolomite, magnesite, serpentine, chlorite, tremolite and anthophyllite as impurities. Many electrical, chemical and laboratory articles are made of steatite and talc. Minor occurrences of poor quality steatite were recorded within Vempalle dolomites near the contact of basic sill near Nagayapalle, 3 km west of Tangedupalle, 1.5 km south of Lingala and 1.5 km west of Rajupalem. A low grade crystalline magnesite body occurs at the base of Vempalle formation, 3 km west of Vempalli town and 2.5 km south east of Kummarampalle (Sunitha V.et al., 2014). Massive steatite when cut into panels is used for switchboards and acid proof tabletops in laboratory, laundry and kitchen sinks, in tubs and tanks as well as for lining alkali tanks in Paper Industry. Due to its high melting point (16300C), soapstone can be used in refractories and fire places. It is also quite useful in sculpturing. Indian talc, especially mined in Rajasthan and Andhra Pradesh, is comparable with the best quality available in other countries. The non pulverised talc is used in refractory. Total reported consumption of talc/steatite/soapstone in the organised sector was at 368 thousand tonnes in 2012-13. About 56% consumption in 2012-13, was in Paper Industry, followed by Paint (20%), Pesticide (11%), Ceramic (8%) and Cosmetic (4%) industries. Nominal consumption was shared by Fertilizer, Rubber, Textile, Chemicals and other industries.

Clay

Superior grade clay useful in ceramic ware and sanitary wares occurs in Hastavaram,Tallapaka and Gadela villages of Rajampeta Mandal and Obulavaripalle Mandal of Kadapa district. The clay deposits are confined to Pullampeta formations of Nallamalai Group of Cuddapah Supergroup. Inferior grade clay occurs in Chinna Orampadu, Anantarajupet, Nandalur, Chitvel and Koduru of Kadapa dist. A variety of clay (Fuller's earth) useful in detergent industries occurs in Venkanarri, Midamala, Chennakesavapuram and Iruvegalapuram along Sagileru River of Kadapa district. Clay and ochre (yellow and red) deposits are found in Vempalle, Narji and Owk Formations at Betamcherla, Ambapuram, Ramallakota, Praema and Palakuru of Kurnool dist. Ochre as fault infillings is noticed near Veldurti of Kurnool district. These clays and ochres can be used in the manufacture of stoneware, sanitary ware, sewer pipes and as filler in paper industry (Source: http://web-grafix.in/apmines/? page_id=889). The total reserves/resources of fuller's earth as per UNFC (United nations framework international) system as on 1.4.2010 are placed at 256.7 million tonnes. Out of these, only 58,200 tonnes are placed under 'reserves' category while about 99.98% are placed under 'resources' category. About 74% resources are located in Rajasthan. The remaining resources are in Andhra Pradesh and other states. Fuller's earth is used usually after activation in bleaching, decolourising vegetable oils, petroleum, lubricants and greases. Recently, the growth in its consumption in this sector has been affected because of advent of more sophisticated techniques in refining and due to availability of effective substitutes like activated bauxite and magnesium silicate. Fuller's earth is generally used in fertilizer industry. Consumption, however, is expected to rise in other unconventional uses as absorbent, cleaning oil spillage on factory floors, as carrier for insecticides, fungicides and as a mineral filler and extender (Indian minerals Year Book: 2013: Fullers Earth).

Iron Ore

The ore minerals from which iron is extracted are hematite (Fe2O3), magnetite (Fe3O4) and goethite (FeOOH). Iron smelting is carried out by reducing iron oxides to iron metal by reaction with carbon monoxide gas, usually derived from coke (Craig et al., 1996). Iron & steel is the driving force behind industrial development in a country. The mining of iron ore, an essential raw material for Iron & Steel Industry is arguably of prime importance among all mining activities undertaken by any country. With the total resources of over 28.52 billion tonnes of haematite (Fe2O3) and magnetite (Fe3O4), India is amongst the leading producers as well as exporters of iron ore in the world. The following types of iron ores are identified in the Nallamalai formations: 1. Micaceous Hematitic ores 2. Banded Hematite Phyllite 3.Banded Hematite Slate 4.Banded Hematite Quartzite 5. Banded Magnetite Quartzite and 6. Hematitic Float Ores (Kasipathi C et al.,2008). Reconnaissance geological surveys conducted in the central and southern parts of Andhra Pradesh for banded iron formations indicated vast resources of lateritic, friable, banded hematitic ores at many places extending to kilometers together in the hill chains and along the foothills. Small deposits of iron ore and hematite occur in the Pulivendula quartzite around Chabali and Pagadalapalle of Pendilimarri Mandal and in Pullampeta quartzite around Mantamampalle of Ontimitta Mandal and Yerraguntla Kota of Koduru

Mandal and 22 km west Rajampeta railway station of Kadapa district. (V.Sunitha et al., 2014). Indian deposits of haematite belong to the Precambrian Iron Ore Series and the ore is within banded iron ore formations occurring as massive, laminated, friable and also in powdery form. Banded Hematite Quartzites are thick sequences of high-grade iron-ores. They are banded, massive and float ore types and occur in parts of Pagadaalapalle (Kadapa District), and Veldurthi – Sarparajupalem-Kalugotla-Ramallakota – Pullagummi – Ghanigattu – Guttupalli – Brahmapuram – R.S.Rangapuram- Muddavaram (Kurnool District). They record rich sequences mostly occupying the Reserve Forest areas. Nearly 3.7 million tonnes of iron ore with 55 to 58% Fe is availble in Kurnool district. The iron ore available in Kurnool district is free from phosphrous and as such it is useful in sponge iron and cement industries. It is proposed to establish a sponge iron industry at Veldurthi. Banded Magnetite Quartzites are found at some places in equally alternate thick bands of magnetite and quartzite. The thickness of magnetite bands vary from minimum 3 m. to 12 m. and that of quartzite is about 8 m. to 15 m. As per UNFC (United nations Framework International) system, the total resources of haematite as on 1.4.2010 are estimated at 17,882 million tonnes of which 8,093 million tonnes (45%) are under 'reserves' category and the balance 9,789 million tonnes (55%) are under 'remaining resources' category. India is the leading producer of iron ore. Iron ore as basic raw material is used for making pig iron, sponge iron and finished steel (Indian Minerals Year Book 2013: Iron).

Dolomite

Dolomite ($CaCO_3.MgCO_3$), in commercial parlance, has 40-45% $MgCO_3$. It is grouped under flux and construction minerals and is important for iron & steel and ferro-alloys industries (Indian minerals Year Book: Dolomite: 2013). Crystalline magnesite is reported from the Vempalle dolomites. This occurrence is confined to the Gulcheru-Vempalle contact and has significance in that it is associated with uranium and phosphorite mineralization (Sinha R.M et al., 1994). Dolomite and calcite are common rock forming minerals. Dolomite is associated with sedimentary carbonate facies and occurs in all geological ages but the economically important deposits are mostly confined to the Precambrian and Palaeozoic eras. Usually the dolomites are associated with limestones and sometimes they occur as irregular beds, lenticles, and pockets and rarely do they occur as hydrothermal vein deposits. The dolostone, loosely called dolomite in most literature, is considered as a principal raw material in the iron and steel industries. It is also used as a refractory material, for which calcined products are preferred. Dolomite is used as a basic lining in open hearth furnaces and in Bessemer converters, for which dead-burnt material is required. Dolomite is also used in the manufacture of high magnesia lime, basic magnesium carbonate, Epsom salts, and for the manufacture of metallic magnesium. Dolomite also finds use in chemical industry, in manufacture of paper, leather, glass etc. and as building material, as terrazzo stucco and also as crushed stones. As per UNFC (United Nations International Framework) System, total reserves and resources of all grades of dolomite are placed at 7533 million tonnes, out of which total reserves are 985 million tonnes and the balance; i.e. 6548 million tonnes are the resources. The share of proved reserves is 5%, probable reserves 8% and remaining resources 87 percent. Of the total resources in India, Andhra Pradesh has 15% (Indian minerals Year Book: 2013).

Kadapa Slabs

Napa slabs/Kadapa slabs are black coloured limestone belonging to Koilakuntla Limestones of Kurnool Group. They occur in the area of Niduzivi,Koduru, Valasapalli areas of Yerraguntla Mandal and Sugumanchupalli areas of Jammalamadugu Mandal of Kadapa district. Due to the development of perfect basal cleavage planes, this rock can be extracted with a thickness ranging from 10mm to 20mm and any size up to 8 feet length. There are about 150 polishing industries in the Kadapa district. These polished slabs are mainly utilized for flooring (Sunitha V et al., 2014).There are about 650 polishing units at Bethamcherla, Dhone, Rangapuram, Kurnool, Owk, Cherlopalli, Ramapuram, Belum, Itikala, Chintayapalli, Ankireddypalli villages/ towns of Kurnool district. Few more polishing units can be established at Bethamcherla. (Source: http://www.kurnool.ap.gov.in/departmentView.apo?mode=getDepartment&departmentFlag=MIG&subDepartmentFlag=MIG).

Possible mineral industries that can be developed in parts of Cuddapah basin

Some of the deposits like limestone, barytes, and dolomite occur extensively over large areas and in huge quantities. Some like graphite, steatite, ball clay, fire clay, china clay, copper, lead, zinc, gold, diamonds, asbestos, iron ore, quartz and silica sand are found in localised pockets, but widely dispersed in different parts of the basin. In addition some small and low grade deposits like corundum, garnet, amethyst, green quartz, feldspar, lime kankar, and lime shell are also being exploited. In addition to these economic and industrial minerals, the basin (and immediate adjacent regions) is endowed with inexhaustible wealth of building stone like marble, slates and different varieties of granites suitable for cutting and polishing. The important minerals produced in the basin are barytes, limestone, asbestos, lead, iron, quartz, dolomite and exportable quality slabs and slates.

Andhra Pradesh State government has given significant number of mining leases and quarry leases for Major minerals (industrial Minerals) and Minor Minerals (Construction Minerals). In addition some Reconnaissance permits (Gold, Diamond, Base metals, Precious metals) have also been given. Mining for Limestone, Barytes, Iron, base metals and Natural Gas falls under large scale mechanized sector.90% of Granite, Dolomite, quartz, feldspar, Clays fall under small sector and remaining 10% falls under medium and large sectors. Tens of principle large and medium scale mineral based industries have been established in Public and Private sectors for manufacture of cement, steel, sponge Iron, fertilizers, Ferro Alloys, Glass, Oil refinery, Fibre Glass, Ceramics, Refractories, zinc refinery, Chemicals etc. Parts of Cuddapah basin of A.P. house the 2nd largest cement producing industries in the country. There are about 5,000 Granite, Marble and Limestone cutting and polishing units, Slate cutting units, Gem cutting and faceting units, Granite monuments manufacturing units, Pulverizing units, Stone crushers, mosaic and ceramic tile units, lime kilns and Rock stone sand units. In addition to strengthen these industries it is essential to give additional support to small scale mineral based industries that can ensure organized exploitation of mineral wealth, without introducing ecological imbalance. AP. Government has come out with a small scale industrial development strategy through establishment of the following types of industries in economically backward segments of Cuddapah basin, especially Kadapa and Kurnool districts. A few suggested industries in Kurnool dist are listed blow, as a sample. Interested can contact A.P. Mines and Geology Department and, A.P. Mineral Development Corporation, for specific details.1)Small scale mosaic tile units 2) ceramic industries3)Cement plants of 100 to 3000 TPD capacity4)Export oriented granite cutting and polishing 5)Pesticide units 6)Small scale calcium carbide & Acetal dehydrate & Acetic acid plants 7). Precipitate Calcium carbonate + Calcium Chloride and Tooth Paste units using the high grade Limestone 8) Dry distemper, Paints9) White and coloured Chalk Crayons units can be setup basing on clay, ochres, and barytes .

Negative Impacts and Initiatives to minimize negative impact on environment:

The environmental impact of mining includes erosion, formation of sinkholes, loss of biodiversity, and contamination of soil, groundwater, surface water by chemicals from mining processes. Dissolution and transport of metals *and heavy metals* by *run-off and ground water is another* example of environmental problems with mining. The implantation of a mine is a major habitat modification, and smaller perturbations occur on a larger scale than exploitation site, mine-waste residuals contamination of the environment for example. Adverse effects can be observed long after the end of the mine activity. Destruction or drastic modification of the original site and anthropogenic substances release can have major impact on biodiversity in the area. Destruction of the habitat is the main component of biodiversity losses, but direct poisoning caused by mine extracted material, and indirect poisoning through food and water can also affect animals, vegetation and microorganisms. Habitat modification such as pH and temperature modification disturb communities in the area. Endemic species are especially sensitive, since they need really specific environmental conditions. Destruction or slight modification of their habitat put them at the risk of extinction. In some cases, additional forest logging is done in the vicinity of mines to increase the available room for the storage of the created debris and soil. Besides creating environmental damage, the contamination resulting from leakage of chemicals also affects the health of the local population. Mining companies in some countries are required to follow environmental and rehabilitation codes, ensuring the area mined is returned to close to its original state. Unfortunately, such measures are not properly followed in remote areas of our country leading to agitations by tribals. Some mining methods may have significant environmental and public health effects. Erosion of exposed hillsides, mine dumps, tailings dams and resultant siltation of drainages, creeks and rivers can significantly impact the surrounding areas (Source: structured using inputs from https://en.wikipedia.org/wiki/Environmental_impact_of_mining).

Where open-pit mining occurs there are not only big holes in the ground, subject to flooding, but also even larger waste disposal areas. Wastes are often dumped in steep sided spoil heaps. The steep slopes are unstable and subject to erosion. Handling and tipping very often result in compaction and loss of soil structure. Some materials may be very fine and left loose and porous. There may be poor drainage or drought. Surface conditions lead to extremes of surface temperature, wind erosion and sand blasting effects. All of these provide in hospitable physical environment for plant growth. These problems must be overcome if the land is to be upgraded from its derelict state and is once again to support healthy environment for habitation. Barytes, Asbestos and Iron mining, using substandard open cast mining practices, especially by private entrepreneurs in the basin created earlier considerable impact on Man, Flora, Fauna and environment. As such there is considerable resentment even for any organized mining practices. This can be changed through socio-economic initiatives.

Since there is a perceptible opposition to mining activities that are affecting adversely both the locals and environment we suggest some prominent approaches to lessen ill effects. It is essential to note that the impact of mineral exploration on the environment depends on such factors as mining procedures, local hydrologic conditions, climate, rock types encountered, size of operation, topography, and many more interrelated factors. Entrepreneurs need to adapt area specific strategies instead of brushing aside the set guidelines as impracticable and irrelevant. They and quality control experts need to take into consideration that exploration and testing stage involves considerably less impact than the mining and processing stages and no precautionary measures be finalized using testing field level pilot plant inputs alone. Mining and processing of mineral resources have a considerable impact on the land, water, air, and biological resources. What must be done, to lessen the impact, is to develop resources with the minimum of adverse impact. However, minimizing environmental degradation caused by mining may be very difficult considering that the demand for minerals is continuing to grow while the volumes of highly concentrated mineral deposits diminish. Therefore, to provide more and more material, larger and larger operations mining ever-poorer grades of ore will be necessary. However, mining of ore deposits that are transitional into the surrounding rocks, allowing recovery of ever-lower grades of ore may need on spot extraction strategies (based on inputs from Edward A.Keller, 1979).

In the recent past a new technique has been developed to avoid pollution of groundwater due to mining waste. Whether the end product is iron or copper, mines generate large quantities of hazardous gunk, known as mine tailings, that has a nasty habit of leaching into groundwater. Now, a new material not too different from ordinary construction concrete is showing promise as an inexpensive and durable means to filter the drippings from pits filled with mine tailings so that only clean water will seep into the ground below. The pervious reactive concrete (PRC), has a pH of 12.5, it is alkaline enough to alter the solubility of heavy metals that come its way. The PRC causes those metals to crystallize and clump together within its concrete pores (Kelleher, S. 2015).

In nutshell the mining activity should be a responsible practice. To ensure completion of reclamation, or restoring mine land for future use, many governments and regulatory authorities around the world require that mining companies post a bond to be held in escrow until productivity of reclaimed land has been convincingly demonstrated. Since 1978 the mining industry has reclaimed more than 2 million acres (8,000 km^2) of land in the United States alone. This reclaimed land has renewed vegetation and wildlife in previous mining lands and can even be used for farming and ranching. These practices in letter and deed need to be implemented to make the mining activity a sustainable developmental initiative in Cuddapah basin in particular and other parts of the country in general. It is essential to note that in many parts of the world, our human footprint has expanded beyond sustainability, and our technologies and habits of exploitation have become excessive leading to irreversible scars on mother earth. In spite of the acceleration of global climate change people can be healthy and productive if we take needed initiatives to safe guard the ecological richness on which we depend. As such, while exploiting mineral wealth we need to ensure that over ambitious exploitation through non-sustainable means is avoided.

Conclusions

The mining industry is a major force in the world economy, occupying a primary position at the start of the resource supply chain. The rapid expansion of mining activity in recent years has made many low and middle income countries more reliant on mining for generating foreign exchange earnings. The rising importance of mining challenges both the critics of mining and its supporters to improve their understanding of the mechanisms that allow mining to have increasingly positive developmental effects. Cuddapah basin, Andhra Pradesh, endowed with rich mineral wealth, is one of the important and fairly well studied geological units in peninsular India. The Middle- Upper Proterozoic Cuddapah basin, has been well known for a variety of mineral resources, such as diamond, barite, asbestos, copper and lead, besides limestone and Cuddapah slabs. Twenty five percent of the world’s barite resources are present within the basin. The metallurgical and mineral industries constitute the bedrock of industrial development as they provide the basic raw materials for most of the industries Barytes, chrysotile asbestos, clays, cement grade limestone and Kadapa slabs/napa slabs are some of the well-known workable mineral deposits associated with Proterozoic sedimentary rocks belonging to Cuddapah Super group and Kurnool Group in Kadapa district. In our presentation we tried to cover other minerals potential too, including Uranium, Diamond, Copper-Zinc-Lead, Gold, Hydrocarbons. We do, however, know this overview has not covered many facets of mineral potential of Cuddapah basin, due to various limitations. However, we do opine the information provided could be used as base information to plan and execute various developmental activities, while taking apt measures to avoid over exploitation of mineral resources that can harm both the Man and the environment.

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